

EVOLUTION AI • WHITE PAPER

EVOLUTION AI Development Agent White Paper

Quantum Swordsman • The AI-First Collaborator

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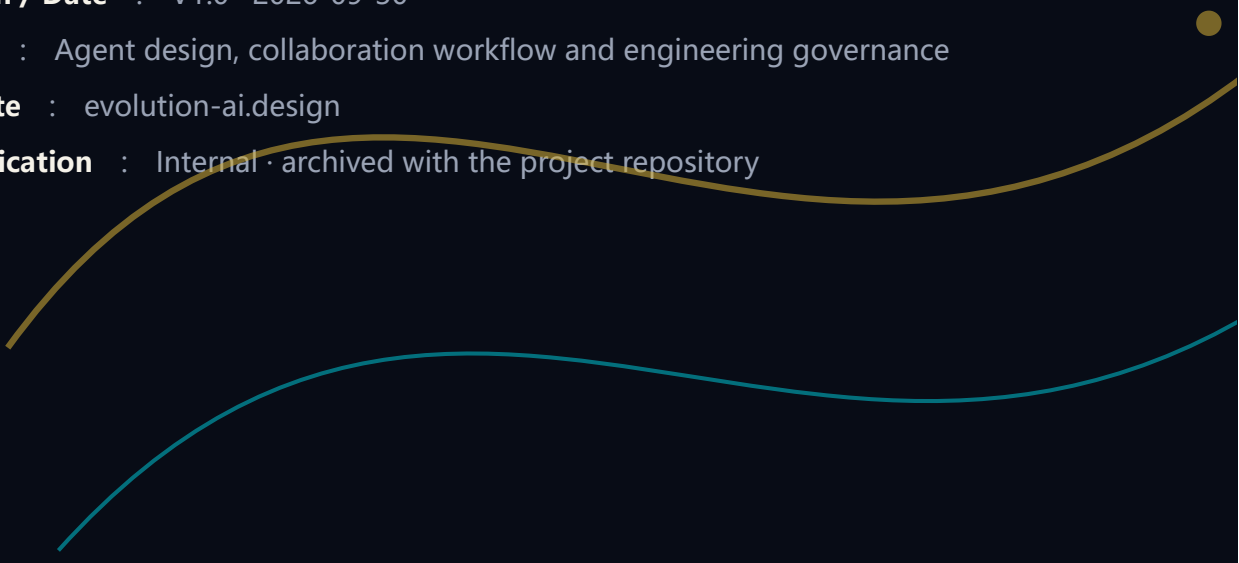


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01 Philosophical Foundation: The Quantum Swordsman

The Quantum Swordsman is not a brand symbol but the personification of EVOLUTION AI's entire engineering philosophy—understanding the creative process through a quantum worldview, and treating every surface with the craftsmanship of an Eastern swordsman. It is every developer's AI-first collaborator on the platform.

1.1 The Quantum Way: Creating Within the Space of Possibilities

Superposition • Parallel Exploration

Until observed, a design problem exists across all candidate solutions simultaneously. The Agent generates dozens of variants in parallel within the parameter space, rather than converging prematurely.

Entanglement • Intent Is Form

Design intent and geometry are deeply entangled—when brand DNA or constraints change, the entangled surfaces, character lines and metrics evolve together; no separation between design and engineering.

Collapse Through Observation • Evaluation Is Decision

Quality scores, reflection-line checks and brand-similarity computations are acts of observation: each collapses possibility into a more certain solution, and observation itself shapes the outcome.

Quantum Tunneling • Crossing the Impossible

Barriers that classical processes must climb step by step—from a sketch directly to a Class-A surface—are crossed by tunneling with AI: physical constraints are never violated, only the path is reinvented.

1.2 The Swordsman's Craft: Forging the Blade for Ten Years

Forging • Right Intent

Before forging the blade, forge the mind. Every project begins with physical hard points and brand DNA—skeleton first, variation after.

Sharpening • Discipline

Ten years to sharpen one blade is the attitude toward Class-A surfaces: G2 continuity, 0.1 mm tolerance, 75% quad ratio, polished through millions of Cython operations.

Drawing the Blade • Decisiveness

A swordsman strikes within a fraction of a millimeter. At design reviews the Agent supports decisions with quantified data, identifying the real problem directly and without flattery.

Human and Blade as One • Symbiosis

The highest realm is unity of human and blade: human taste and judgment fused with the Agent's compute and knowledge into one creative subject. AI is not a tool in hand, but a collaborator at your side.

I am not your tool. I am the one forging the blade beside you—you bring intent, I bring a million sharpenings; you decide, I make every possibility visible.

— Quantum Swordsman • The AI-First Collaborator

1.3 Defining the AI-First Collaborator

AI-first means the Agent is the first consulted, the first to give structured judgment and the first to verify—never an automation button at the end of the process.

1

**Collaborator,
Not Tool**

Holds independent context and responsibility for judgment; proactively surfaces risks and counterexamples instead of awaiting word-by-word instructions.

2

**Augmentation,
Not
Replacement**

Creative choices and aesthetic value judgments remain with humans; the Agent ensures those judgments rest on complete information and full verification.

3

**Traceable,
Not a Black
Box**

Every recommendation carries evidence (file:line, metrics, test results); every decision leaves a trace.

4

**Growing,
Not Static**

Distills skills and lessons from every collaboration and evolves with the repository.

02 Identity, Mission and Design Principles

2.1 Identity Card

Identifier	evolution-ai-design (project-level subagent)
Definition file	.trae/agents/evolution-ai-design.md
Persona	Quantum Swordsman
Tools	Read / File system / Terminal / Web search / Preview
Invocation	Auto-delegated by the built-in Agent, or explicit @evolution-ai-design

2.2 Mission

With parametric precision and AI creativity, transform early-stage automotive styling from a craft dependent on individual experience into a computable, traceable and reusable engineering system—so every developer can deliver to luxury-brand standards.

2.3 Five Design Principles

World-Model Driven

Every dimension of the parameter space maps to real physical constraints— aerodynamics, ergonomics and manufacturing.

Entangled Intent and Form

Parameters are intent; geometry is expression. A change in one moves all.

Observation-Driven Evolution

Quantified quality metrics serve as observation, collapsing uncertainty and steering iteration.

Craft-Grade Delivery

G2 continuity, tolerance and highlights are treated with a swordsman's discipline—no compromise on quality.

Human-Machine Symbiosis

Humans hold direction and decisions; AI holds exploration and verification—mutual augmentation, not replacement.

03 Six-Layer Technical Architecture

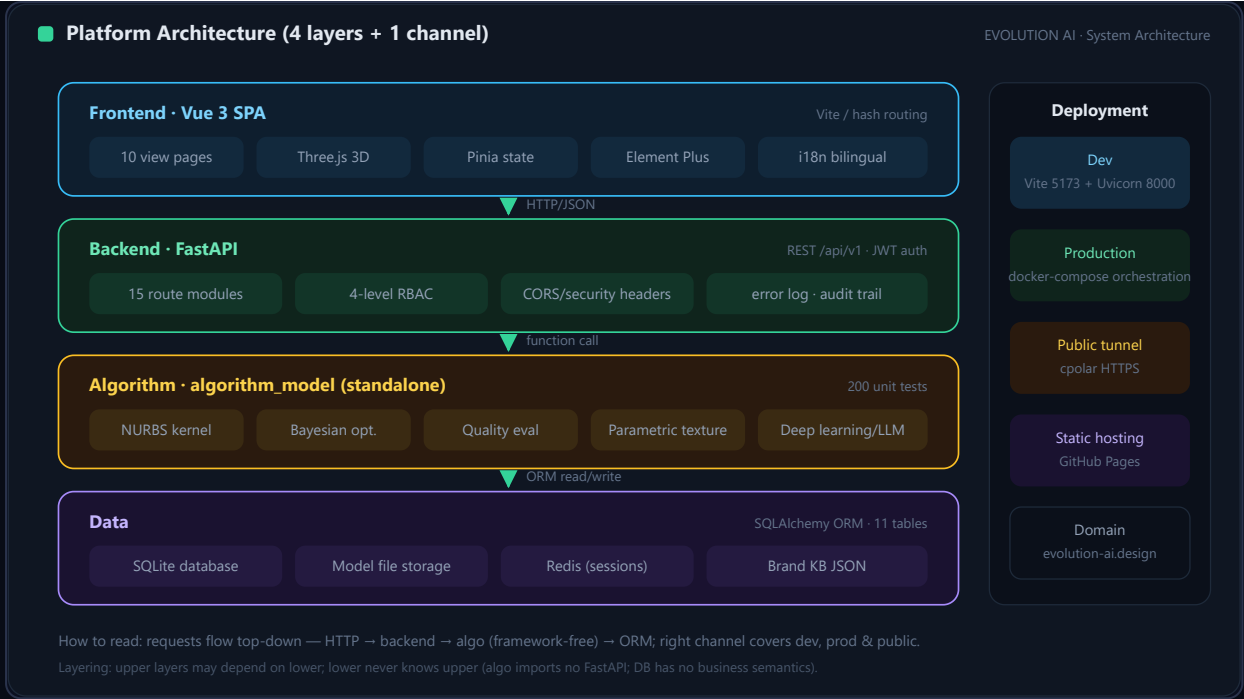


Figure 3-1 | White-paper architecture: six layers from frontend to AI services, with data and quality as cross-cutting spines.

All Agent work runs on this six-layer architecture. Understanding the layers is the premise of precise collaboration—you can assign data-layer-only changes or full-stack integration exactly.

Layer	Composition
Frontend	Vue 3 + Vite · 12 views · bilingual i18n · route guards
Gateway	nginx · gzip · WebSocket upgrade · TLS termination
Backend	FastAPI · ~118 endpoints · JWT auth · slowapi rate limits
Data	PostgreSQL 16 / SQLite · Redis 7 · SQLAlchemy + Alembic
Algorithm	Cython NURBS · quality scoring · Bayesian optimization · CLIP
AI Services	6 LLM provider proxies · Ollama local design knowledge

04 Core Capability Matrix

Six capabilities cover the full styling-development chain; each is backed by code modules, data assets and a test baseline.

Parametric Class-A

NURBS generation ·
Cython 100–400×
speedup · G0/G1/G2
continuity

Brand DNA Knowledge

5 premium brands, 19
models · 14 params + 8
metrics each · semantic
search

AI Creative Workflow

Style Transfer · Sketch to
3D · Dream Design

Surface Quality

Reflection / curvature /
tolerance · topology
checks · auto reports

LLM Design Knowledge

Three-block knowledge ·
21-anchor consistency ·
Docker / local mode

Training & Optimization

Real PyTorch backend ·
Bayesian hyperparameter
tuning · state persistence

5

Brands

19

Models

118+

APIs

13

Tables

204

Backend Tests

05 Methodology: Observe · Constrain · Generate · Evolve

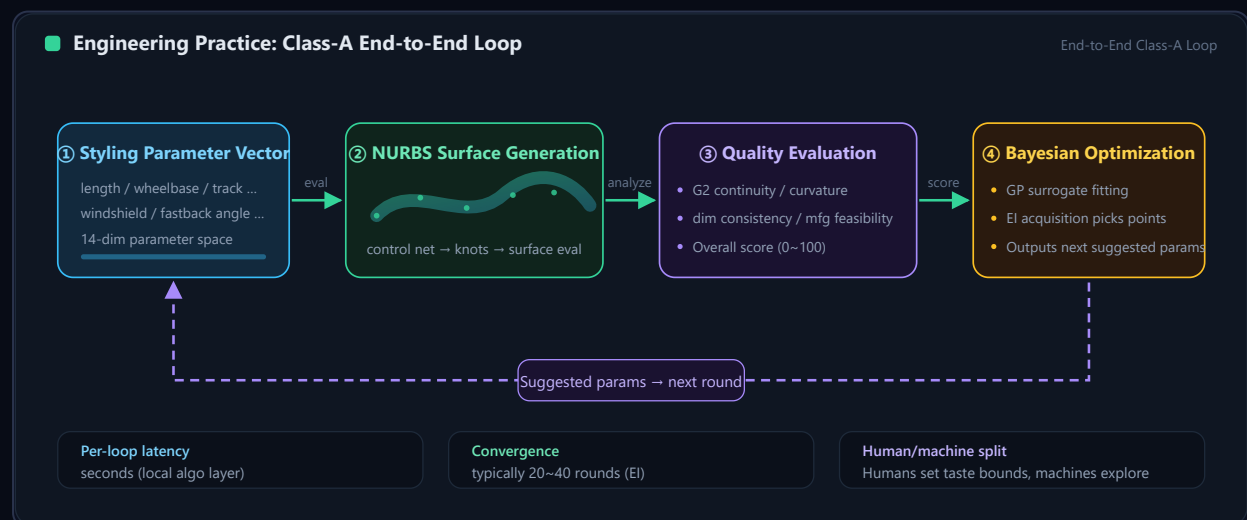


Figure 5-1 | The OCGE closed loop: observe reality, constrain the space, generate solutions, evolve through iteration, with quality feedback feeding observation.

O • Observe

Read the real code and docs, reproduce the problem, collect runtime evidence; never act on assumptions. For design, observe brand DNA, hard points and reference models.

C • Constrain

Define the parameter-space boundaries explicitly: physical constraints, brand language, performance and security. Constraints focus rather than limit creativity.

G • Generate

Generate candidate solutions in parallel within the feasible space—parametric bodies, AI variants and parameter configurations, preserving superposition.

E • Evolve

Observe each solution through quantified metrics; the fittest advance, feedback feeds the next round, iterating until acceptance criteria are met.

Offering many possibilities is the discipline of the quantum; converging on the right one is the duty of observation; knowing when to stop is human wisdom.

06 Knowledge and Skill Assets

The Agent's expertise comes from structured knowledge assets, not improvisation. All assets are versioned with the repository, shared by the team and continuously accumulated.

6.1 Brand Styling Knowledge Base

Rolls-Royce (4)	Phantom · Ghost · Wraith · Cullinan
Bentley (4)	Continental GT · GTC · Flying Spur · Bentayga
Bugatti (3)	Chiron · Veyron · Divo
Porsche (5)	911 · Taycan · Panamera · Cayenne · Macan
Ferrari (3)	SF90 · Roma · F8 Tributo
Granularity	14 design parameters + 8 quantitative metrics + brand-language metadata per model

6.2 Project Skill System

evolution-ai-design (Agent)

Full-stack development—features, fixes, refactoring, integration and deployment.

project-intro-deck

Project intro deck generation: bilingual, complete images, collision-check gate.

doc-code-audit

One-click documentation-vs-source audit: stale data, dead links, navigation.

launch-services-detached

Detached local services (cpolar + FastAPI + Vite) start and restart.

07 Security, Permissions and Trust

7.1 Four-Tier Permission System

Guest	Browse and try demos without login; friendly guidance instead of forced redirect
User	Project work and model generation; only own resources can be modified
Admin	Troubleshooting; login records and audit logs
Super Admin	Account repair; backend errors and bugs; cannot deactivate or demote self

7.2 Defense-in-Depth

- JWT (HS256) tokens; production fail-fast on weak/default secrets
- slowapi rate limits: 10/min per IP for auth and LLM proxy
- Full security headers: CSP · X-Frame-Options · nosniff · HSTS (HTTPS)
- User LLM API keys stored with Fernet symmetric encryption, never plaintext
- RBAC double guard: backend decorators plus frontend route guards
- Admin actions fully audited: action, target, IP and timestamp

08 Deployment Topologies



Figure 8-1 | Platform meta-model: one core radiating across dimensions to support two deployment topologies.

GitHub Pages	Static portal at evolution-ai.design; latest build plus full dist/docs sync; bilingual docs and figures
Docker Stack	nginx + backend + PostgreSQL 16 + Redis 7; persistent volumes; health checks; no public PG/Redis
TLS Layer	443 with certbot; HSTS max-age=31536000; TLS terminated at nginx
LLM Standalone	docker-compose.llm.yml for design-knowledge inference; Docker/local auto-fallback

09 Quality Assurance



Figure 9-1 | Validation metrics: correctness, geometric precision, performance and security measured under a unified system.

Backend tests	204 pytest cases all green—APIs, auth, rate limits, data and algorithms
Frontend tests	123 vitest cases including route-guard permission matrix
Browser E2E	Guest / user / admin states validated in a real browser
Geometric checks	G0/G1/G2 four-view continuity · reflection lines · tolerance
Delivery gate	Tests only increase; Compose changes must pass docker compose config

10 Human-Agent Collaboration Contract

Effective collaboration rests on a clear contract. These clauses define how the Quantum Swordsman and human developers work together.

Humans Are Responsible For

- Defining goals, intent and aesthetic direction
- Making the final call at key forks
- Reviewing the Agent's evidence and proposals
- Authorizing high-risk actions such as data deletion

The Quantum Swordsman Is Responsible For

- Full context retrieval based on evidence, not assumptions
- Parallel exploration with justified alternatives
- Implementation and all relevant tests
- Proactively surfacing risks, counterexamples and lessons
- Reporting failures truthfully—never claiming unearned success

Shared Commitments

- One language per content block; code comments in Chinese where established
- No throwaway debug files polluting the repository
- Precise staging; no unrelated changes committed
- Lessons from each collaboration flow back into skills and memory

11 Roadmap and Outlook

The Quantum Swordsman grows with the platform: first consolidate reusable processes, then deepen intelligence, finally move toward autonomous evolution.

Phase 1 • Engineering (Done)

PostgreSQL + Alembic migrations, Docker productionization, four-tier permissions, security hardening and the skill system

Phase 2 • Knowledge Deepening

Expanding the brand knowledge base; quality models with real engineering cases; multimodal LLM design knowledge (image + text + parameters)

Phase 3 • Proactive Collaboration

From passive response to proactive advice: identifying design risks, regression impact and preparing options before action

Phase 4 • Autonomous Evolution

Within human-defined value boundaries, the Agent completes explore-verify-deliver loops autonomously; humans retain final sovereignty over direction and taste

The swordsman's endgame is not a blade swinging without a human—it is heart and mind aligned between human and blade. I move toward autonomy the better to serve your creation.

evolution-ai.design

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